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CRIME DATA ANALYSIS

Project Documentation

Tool Used: Microsoft Excel

Data Analysis, Cleaning, PivotTables, PivotCharts and Dashboard

1. Introduction

This project analyzes crime data using Microsoft Excel. The main purpose of the project was to clean the crime dataset, analyze crime incidents, and present the results using PivotTables, PivotCharts, and Dashboards. Excel formulas and data-cleaning techniques were also used to identify important patterns related to crime types, districts, case status, victim gender, severity level, and crime trends.

2. Project Objectives

- Clean the crime dataset.
- Identify and remove duplicate records.
- Handle missing and invalid values.
- Standardize inconsistent data.
- Perform data analysis using Excel formulas.
- Create PivotTables and PivotCharts.
- Design a crime analysis dashboard.
- Identify important crime patterns and trends.
- Provide useful findings and recommendations.

3. Dataset Description

The dataset contains information about reported crime incidents.

Original Records: 300

Duplicate Records Removed: 8

Final Cleaned Records: 292

The dataset includes:

- Crime ID
- Crime Date
- Crime Type
- District
- Victim Gender
- Case Status
- Severity Level

4. Problems Found

Before performing the analysis, the dataset was carefully inspected to identify data-quality problems. Several missing values and duplicate records were found. These issues could affect the accuracy of the analysis, PivotTables, charts, and dashboard results.

4.1 Duplicate Records

The original dataset contained 300 records. During the inspection process, 8 duplicate records were identified and removed. After removing the duplicates, the final dataset contained 292 records for analysis.

4.2 Missing Values

A total of 142 missing values were identified across the main columns of the dataset.

Column	Missing Rows
ID	15
Crime_Date	17
Crime_Type	15
District	16
Gender	45
Case_Status	16
Severity_Level	18
Total Missing Rows	142

4.3 Missing ID

There were 15 missing values in the ID column. Since each crime record should have an identifier, these missing values were reviewed and handled during the data-cleaning process.

4.4 Missing Crime Date

The Crime_Date column contained 17 missing values. Missing dates can affect time-based analysis, such as monthly and yearly crime trends.

4.5 Missing Crime Type

There were 15 missing Crime_Type values. Crime Type is important for identifying the most common types of crimes and comparing crime categories.

4.6 Missing District

The District column had 16 missing values. This can affect geographical analysis and make it difficult to determine which districts have higher crime levels.

4.7 Missing Gender

The Gender column had the highest number of missing values, with 45 missing rows. This can affect analysis of crime incidents based on victim gender.

4.8 Missing Case Status

There were 16 missing values in the Case_Status column. Case Status is important for determining whether cases are closed, open, referred, or under investigation.

4.9 Missing Severity Level

The Severity_Level column contained 18 missing values. Missing severity information can affect the analysis of low-, medium-, and high-severity crimes.

Summary of Problems Found

- 8 duplicate records
- 15 missing ID values
- 17 missing Crime_Date values
- 15 missing Crime_Type values
- 16 missing District values

- 45 missing Gender values
- 16 missing Case_Status values
- 18 missing Severity_Level values
- Invalid or inconsistent entries and formatting

5. Data Cleaning Process

- Removed 8 duplicate records.
- Identified a total of 142 missing values across the dataset.
- Reviewed the 15 missing ID values.
- Reviewed the 17 missing Crime_Date values.
- Handled the 15 missing Crime_Type values.
- Handled the 16 missing District values.
- Reviewed the 45 missing Gender values.
- Handled the 16 missing Case_Status values.
- Reviewed the 18 missing Severity_Level values.
- Standardized inconsistent text entries.
- Checked the cleaned dataset for consistency before analysis.

After removing the 8 duplicate records, the final dataset contained 292 records.

6. Excel Formulas and Functions Used

Formula / Function	Purpose
COUNTA	Count total crime records
COUNTIF	Count crimes according to categories
COUNTIFS	Count records based on multiple conditions
TRIM	Remove extra spaces
PROPER	Standardize text capitalization
UPPER	Convert text to uppercase
LOWER	Convert text to lowercase
YEAR	Extract crime year from Crime Date
MONTH	Extract crime month
IF	Create data-quality and status conditions
PivotTable	Summarize crime data
PivotChart	Visualize crime patterns

7. Data Analysis

The cleaned dataset was analyzed using Excel formulas, PivotTables, PivotCharts, and the dashboard.

Metric	Result
Total Crime Incidents	292
Most Common Crime Type	Theft – 85
Highest Crime District	Hodan – 84
High Severity Incidents	106
Medium Severity Incidents	156
Low Severity Incidents	30

Closed Cases	123
Open Cases	67
Under Investigation	77
Referred Cases	25
Male Victims	160
Female Victims	89
Unknown Gender	43

Crime Type Analysis

The PivotTable shows that Theft was the most common crime type, with 85 incidents.

- Theft – 85 incidents
- Assault – 60 incidents
- Fraud – 57 incidents
- Robbery – 44 incidents
- Traffic Offense – 28 incidents
- Burglary – 18 incidents

District Analysis

Hodan recorded the highest number of reported crimes with 84 incidents.

- Wadajir – 62
- Yaqshid – 44
- Daynile – 38
- Karan – 32
- Waberi – 32

Case Status Analysis

- 123 Closed cases
- 67 Open cases
- 77 Under Investigation
- 25 Referred cases

This indicates that a considerable number of cases remain active or under investigation.

Severity Analysis

- Medium – 156 incidents
- High – 106 incidents
- Low – 30 incidents

High-severity incidents represent a significant portion of the total reported crimes.

8. Key Findings

- Hodan has the highest number of reported crimes, with 84 incidents.
- Theft is the most common crime type, with 85 incidents.
- Medium-severity incidents are the largest category, with 156 cases, while 106 incidents are classified as high severity.
- Closed cases make up a substantial portion of the dataset, with 123 cases.
- Many cases remain active or under investigation, with 67 Open cases and 77 Under Investigation cases.
- Male victims represent the largest recorded gender group, with 160 incidents.
- The dataset contained significant missing and unknown information that required cleaning before analysis.
- The dashboard and PivotTables provide a clear way to monitor crime patterns by district, crime type, status, severity, gender, and time.

9. Strategic Recommendations

- Increase police resources in Hodan and Wadajir, since these districts recorded high numbers of reported crimes.
- Strengthen theft prevention through public awareness, surveillance, and community policing.
- Create a case-clearance team to focus on Open and Under Investigation cases.
- Improve data quality by reducing missing and unknown values in crime records.
- Improve coordination between police and other relevant authorities for Referred cases.
- Monitor high-severity incidents and prioritize resources toward serious crimes.
- Continue using the Excel dashboard to monitor crime trends and support evidence-based decision-making.

10. Conclusion

This project successfully cleaned, analyzed, and visualized crime data using Microsoft Excel. Different Excel tools, including formulas, PivotTables, PivotCharts, and Dashboards, were used to transform raw crime data into useful information.

The analysis identified important patterns, including the high number of crimes in Hodan, the high frequency of Theft, the significant number of High and Medium severity cases, and the large number of cases that are Closed, Open, or Under Investigation.

Overall, the project improved the quality of the crime dataset and produced useful insights that can help police departments and relevant authorities make better decisions about crime prevention, resource allocation, case management, and public safety.

Data Quality Summary

Original Records: 300

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Total Missing Values Identified: 142